

AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_INDEX

Submission: model_8

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution sets up the correct framework — the polynomial encoding and the cyclotomic factorisations — and evidently has the right ideas, but throughout it reads as a sketch rather than a proof: the decisive steps are asserted, not justified. Most importantly, it does not prove that the index cannot be ≥ 6 : it merely states that “quotient non-negativity” forces the degree above 19, without deriving the coefficient constraint or exhibiting the offending negative coefficient (**NEC1/NEC2**). The claim $N = 5$ is likewise asserted, and the lower-bound argument for the minimal configuration is not carried out — e.g. the case $A(1) = 2$ is never excluded, so the required size bound is not established (**NEC5**). Because even the proof that the index is ≤ 6 lacks enough detail to stand, the submission does not reach the essentially-complete threshold and does not merit the partial-progress mark either. It therefore scores **0**.